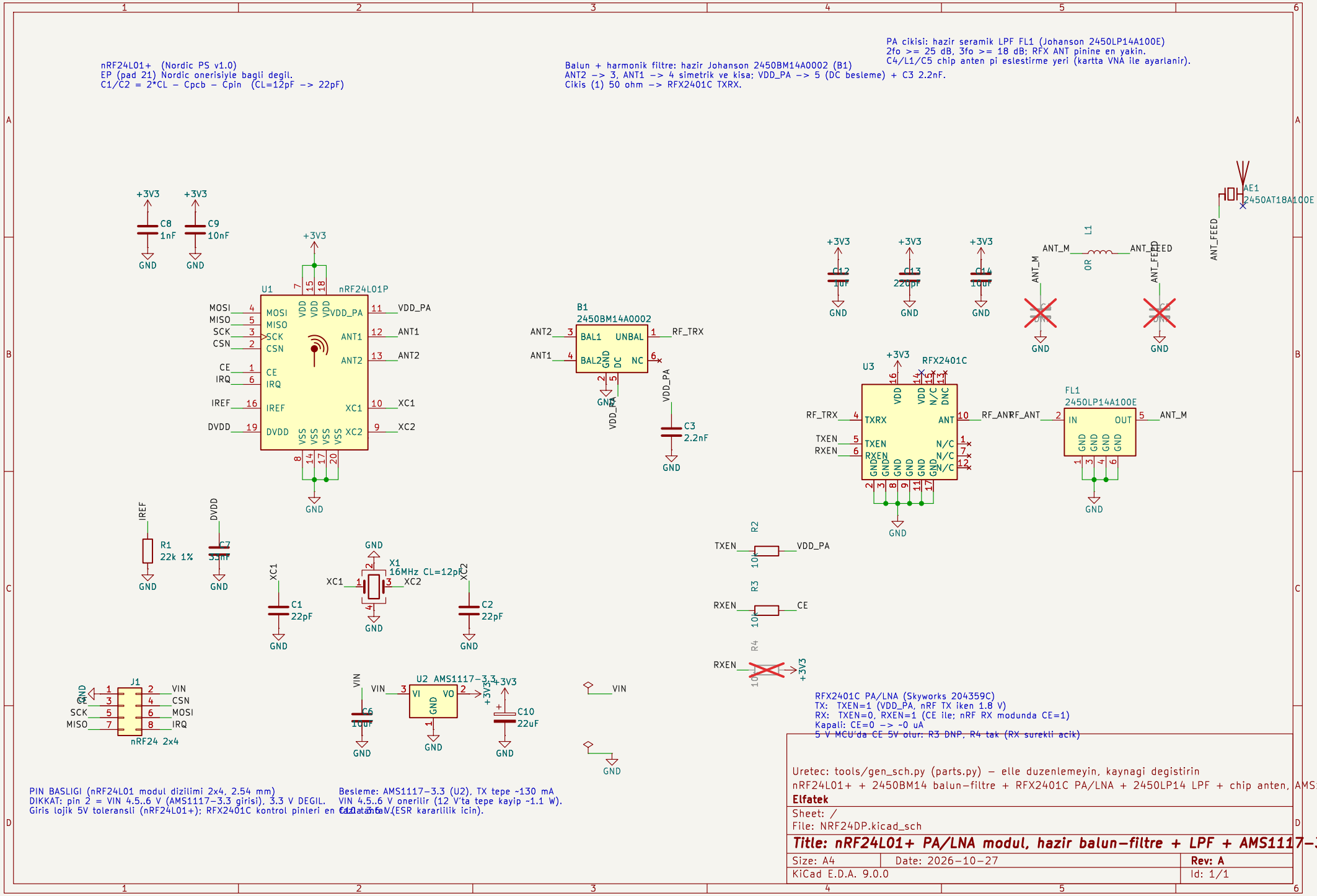

nRF24L01+ (Nordic PS v1.0)
EP (pad 21) Nordic onerisiyle bagli degil.
C1/C2 = 2*CL − Cpcb − Cpin (CL=12pF −> 22pF)
Balun + harmonik filtre: hazir Johanson 2450BM14A0002 (B1)
ANT2 −> 3, ANT1 −> 4 simetrik ve kisa; VDD_PA −> 5 (DC besleme) + C3 2.2nF.
Cikis (1) 50 ohm −> RFX2401C TXRX.
PA cikisi: hazir seramik LPF FL1 (Johanson 2450LP14A100E)
2fo >= 25 dB, 3fo >= 18 dB; RFX ANT pinine en yakin.
C4/L1/C5 chip anten pi eslestirme yeri (kartta VNA ile ayarlanir).
U1
nRF24L01P
B1
2450BM14A0002
U3
RFX2401C
FL1
2450LP14A100E
AE1
2450AT18A100E
U2 AMS1117-3.3
J1
nRF24 2x4
RFX2401C PA/LNA (Skyworks 204359C)
TX: TXEN=1 (VDD_PA, nRF TX iken 1.8 V)
RX: TXEN=0, RXEN=1 (CE ile; nRF RX modunda CE=1)
Kapali: CE=0 −> ~0 uA
5 V MCU'da CE 5V olur: R3 DNP, R4 tak (RX surekli acik)
PIN BASLIGI (nRF24L01 modul dizilimi 2x4, 2.54 mm)
DIKKAT: pin 2 = VIN 4.5..6 V (AMS1117-3.3 girisi), 3.3 V DEGIL.
Giris lojik 5V toleransli (nRF24L01+): RFX2401C kontrol pinleri en fazla 3.6 V.
Besleme: AMS1117-3.3 (U2), TX tepe ~130 mA
VIN 4.5..6 V onerilir (12 V'ta tepe kayip ~1.1 W).
C10 tantal (ESR kararlilik icin).
Uretec: tools/gen_sch.py (parts.py) − elle duzenlemeyin, kaynagi degistirin
nRF24L01+ + 2450BM14 balun−filtre + RFX2401C PA/LNA + 2450LP14 LPF + chip anten, AMS1117−
Elfatek
Sheet: /
File: NRF24DP.kicad_sch
Title: nRF24L01+ PA/LNA modul, hazir balun−filtre + LPF + AMS1117−3.3,
Size: A4
Date: 2026-10-27
Rev: A
KiCad E.D.A. 9.0.0
Id: 1/1